



L1a: Course Orientation, Markets, Assets, and the Decision Problem

CHEME 5660 Cornell University

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Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we locate the course, argue that you are **already a participant** in the markets it studies, and end on a question this lecture will not answer.

Today's concepts

- **Locate the course:** how four units connect cash-flow models, uncertainty, contingent claims, and algorithmic decisions.
- **Why you are already in this:** retirement accounts make nearly every working professional a portfolio holder, and the markets involved are roughly a thousand times the daily scale of the industries this college trains you to enter.
- **The decision problem:** an instrument is a schedule of dated payments, and two schedules cannot be ranked without a rule for moving dollars across time.

Note: this lecture has no notebook and no notes packet. These slides are the record.

Summary

This lecture located the course, argued that you are already a market participant, and read five financial promises as schedules of dated payments.

Key takeaways

- **You are already a portfolio holder.** Retirement accounts make market participation the default, so the allocation question is one nearly everyone answers, most of them by inaction.
- **An instrument is a schedule.** A bill, a note, a bond, a corporate bond, and an inflation-indexed security differ in their dated cash flows, not in kind: the product names are labels for those schedules.
- **A claim is only as good as the rung it survived.** Theory, backtest, forward test, and real money each see different data, and each is the only stage able to falsify its own claim.

On Thursday we build the rule that lets us rank those five schedules.